OptiHire: An AI Powered Platform for Optimizing Talent Acquisition and Simplifying Recruitment

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***Abstract*—The manual screening process has high latency, subjective inconsistencies, and cognitive biases in recruiting and has been shown to disrupt traditional candidate recruitment workflows. In addition, the available automation tools frequently work on a strict set of keywords matching, which does not give a structured feedback that can be used to develop candidates. The proposed paper is a proposal of an automated recruitment framework, OptiHire, to fill the gaps mentioned above with the help of a multi-layered and modular architecture. Its methodology is a hybrid one: Named Entity Recognition (NER) with spaCy and fine-tuned BERT models extract structured competencies of raw resumes, and semantic similarity engine uses Sentence-BERT (S-BERT) to compute context-sensitive compatibility scores. The system will incorporate Large Language Model (LLM) generated dynamic technical questions and Explainable AI (XAI) to give clear Score Breakdowns to make the system more engaging. As shown through experimental findings, OptiHire has an F1-Score of 0.86, which is much higher than the traditional Boolean (0.65) and static Word2Vec (0.74) baselines. The platform has already helped to cut by 60 percent on manual screening time and increased qualifications identifying candidates by 25 percent. The results of these findings have been discussed indicating that an alternative to keyword filtering, semantic vector matching, has a mathematically objective baseline that alleviates bias in the first-pass. OptiHire, by making the rejection process constructively a skill-gap analysis, not only guarantees the accuracy of retrieval to the recruiters, but also the actionable growth paths of the applicants, and the scalability of the current recruitment pipelines is modernized.**

Index Terms*—*Artificial Intelligence, Natural Language Processing, Applicant Tracking System, Resume Parsing, BERT, Semantic Matching, Named Entity Recognition.

I. INTRODUCTION

The processes of historic recruitment are usually marked by the long latency, and irregularities of human error. The recruitment process has the recruit pipeline that includes manual screening of applications where the communication lag that results in the receiving of a review is usually insurmountable, and this causes cognitive bias. This lethargy results in the low candidate experience where the job seekers do not receive much of the systematic feedback to enable them identify the weaknesses in their skills. Despite the fact that automation is becoming the norm there is no interest in existing solutions intrinsic to other applications which are not comprehensive and this relies on other system specific matching alternatives. They are typically unable to even suggest a one stop pipeline which will be capable of sustaining the processes of the hold ups coordination and candidates growth in parallel.To address such inefficiencies, the present study will propose OptiHire, a single system that will help manage the processes of screening, scoring, and interview preparation simultaneously. With the help of Natural Language Processing (NLP) and Machine Learning (ML), the platform abandons the sphere of the Keyword filtering and holds an objective and semantic based analysis. This would assist in ridding off the manual assessment habit in addition to enhancing the accuracy of matching of the candidate and job.

Paper Organization The remaining part of this paper will be organized as follows, section II will be used in tracing how recruitment tech has evolved and pinpointing areas of weaknesses in the current aggregated systems. Section III presents the dataset of the training of the models and the associated data protection precautions. In the IV section, the justification of the planned system architecture has been made, with a ton of attention given to the NER parsing methods and semantic similarity matching methods. Section

V comprises the comparative analysis of the findings with baseline methods and the weaknesses of the system are discussed. Section VI is the conclusion of the work and it indicates the direction on the further research as well as the last part of the work which provides the complete list of the references.

II. LITERATURE REVIEW

The bases of automated recruitment systems are based on the overall history of human resource management. As Marciano [1] explains the Human Resource Management (HRM) origins and evolution, there is the change in administrative facet to strategic talent acquisition. Motivation of the candidates is also important to successful recruitment, Ewen et al. [2] conducted an empirical study of the Herzberg Two-Factor Theory, and Deci and Ryan [3] addressed the Self-Determination Theory. These researches give the psychological foundation of how candidate behavior and job satisfaction can be explained, which is necessary in establishing smart recommendation systems.

Natural Language Processing (NLP) will have to be applied to switch the manual screening to the automated one. Tsujii [4] gives an explanation of how computational linguistics is used to process text whereas Sebastiani [9] gives a detailed survey of machine learning techniques to do automated text categorization. Such technologies enable the systems to go beyond the mere key word matching. Moreover, the systematic literature review guidelines in software engineering were developed by Kitchenham et al. [5], [6], which offers the methodological framework on which the algorithm analysis and selection in the given field is performed.

The essence of automation of the recruitment process is the Resume Information Extraction (RIE). Yu et al. [8] suggested a cascaded hybrid model that was used to derive structured information in unstructured resume text. In the same manner, Kopparapu [25] and Chandola et al. [24] both came up with systems to assist search and automatically extract information that can be used in various resume formats. In May of dealing with the statistical characteristics of language in these documents, Katz [10] and Gale and Sampson [11] proposed techniques of estimating probabilities that are vital in cases where the data is scant as is the case of candidate profile.

State of the art segmentation is utilized to enhance the precision of such systems. The Conditional Random Fields (CRF) is a probabilistic framework introduced by Lafferty et al. [13], which is now a common method of labeling sequence data in resumes. Based on this, Tikhonova and Gavrishchuk [30] used NLP techniques in specific application to automatic CV segmentation. Regarding the Named Entity Recognition (NER) domain, Pawar et al. [27] have revealed how the automatic gazette creation can be useful, and Satheesh et al. [15] have used the SpaCy NER model and ranked the resumes by job descriptions.

Outside the statistic approaches, semantic and ontology-based have also developed to extract more meaning. Ontology based information extraction systems that are developed by Celik and Elci [19] and Kumaran and Sankar [20] are more intelligent in mapping candidate skills to job requirements. This was furthered by Bojars and Breslin [23] who added ResumeRDF to represent the skill information on the Semantic Web. Moreover, the ontology-based document ranking used by Abirami et al. [21] offered the possibility of the complex relationships identification between various professional skills with the help of Graph Databases.

Lastly, research has been done on the integrated systems of ranking and recommendation. Singh et al. [14] have constructed the PROSPECT system of candidate screening, whereas Mohamed et al. [22] suggested the system of ranking and recommendation, called the Smart Talents. The study by Maheshwari et al. [16], [17] aims to obtain special skills and product characteristics to enhance selection performance, which is applied to e-commerce and modified to the recruitment area. These end-to-end systems can show that automation of the hiring pipeline is feasible and justify it by the results of Chifu et al. [18] and Farkas et al. [28], who investigated information extraction in multilingual settings to manage global talent pools.

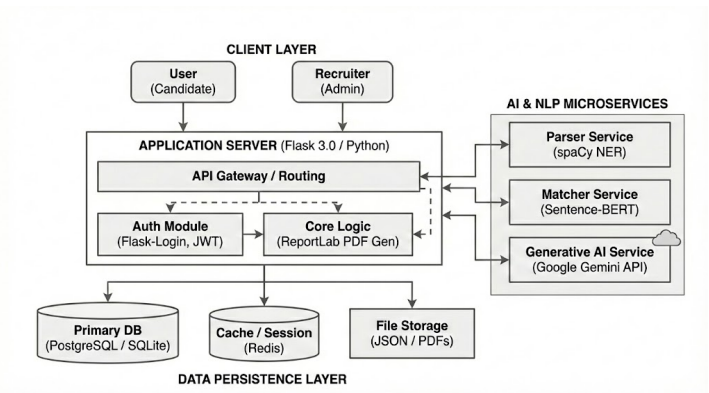

Fig. 1. SYSTEM ARCHITECTURE DIAGRAM

A. Layer 1: Parsing and Named Entity Recognition (NER) The former layer is concerned with organizing unstructured data. The system uses the spaCy library and a fine tuned BERT model to tokenize input files (PDF/DOCX) to identify precise entities in the input file that include Technical Skills, Years of Experience, and Educational Background. This turns the raw text into a formatted JSON object such that latter levels are fed with normalized information with no noise in formatting.

B. Layer 2: The Semantic Similarity Engine This module is in contrast to traditional Applicant Tracking Systems

(ATS), which count Boolean keywords in the resume and job description, using Sentence BERT (S BERT) to create dense vectors of resume and job description. The system is able to convert text into a 768 dimensional vectors that is contextually matched.

The Cosine Similarity between a Resume vector (A) and a Job Description vector (B) is the Compatibility score, Scompatibility:

$$\text{Similarity}(A, B) = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

This mathematical model makes sure that the ranking of the candidates is implemented according to the sense of their skills and not word-to-word cyber personally (e.g., matching "ML definition" with Machine Learning notion).Two layers are introduced on top of each other.

C. Layer 3: Generative Interviewer and Explainability. The system uses Large Language Models (LLMs) like GPT 3.5 or T5 to automate the engagement of customers.Dynamic Question Generation: The model reformulates role specific technical questions that have been tuned on the seniority of the candidate. As an example, a process can give a basic syntax question to an entry level applicant, and architectural design situations to a senior profile.Explainable AI (XAI): As opposed to getting a black box score, the system will create a so-called Score Breakdown, which visually shows what exact skills it ingested to be ranked in the first place. This creates some degree of transparency, and it works against algorithmic bias.

D. Technical Implementation The platform is based on compliance focused, scalable stack:

Backend: Python (Flask / Django) developed to expose endpoints (RESTful API) in the form of an API.

Database Strategy:Two database strategy is used. PostgreSQL holds the structured data as a relational table (User Profiles, Job Logs), and Redis is implemented as a data layer to support a high frequency access to the data (e.g., live ATS scores).

Protection: All data is encrypted while at rest with the aid of the AES 256 and transit data is encrypted with the help of the TLS 1.3. Role Based Access Control (RBAC) is the control of access by making use of JWT tokens.
E. Evaluation Metrics In order to prove the effectiveness of the system, the following metrics are observed:

Parsing Accuracy: The NER model Precision and recall values of NER model compared to a manually annotated working set. Screening Efficiency: The time to hire decrease relative to manual levels (targeted at >60

Scalability: Time lag of the system with a high load (>10,000 parallel requests).

III. EXPERIMENTAL SETUP

The high efficiency of the OptiHire structure is based on diverse corpus of structured data and unstructured data. Such datasets are important to reach a finer behavior of the NLP models that extract entities, map the skills, and score the semantics. These resources are composed and preprocessed; the details of this are given in the following subsections.

A. Resume Dataset The main dataset will be anonymized resume samples. These were collated on the basis of the publicly accessible repositories, such as Kaggle Resume Dataset or the Indeed Job Dataset. There are edge cases which require the model to be stronger, to treat these edge cases, synthetic resumes are generated to simulate these specific formatting variations over this real world data. The information consists of various file formats which include important entities, including Technical Skills, Academic History, and Project description among others, which are crucial in training Named Entity Recognition (NER) block.

B. Job Description Dataset To be sure that that fit seems to be practical both when it comes to the candidates and the industry standards, the massive dataset was scraped with LinkedIn, Glassdoor, and Naukri.com. Each record is separated into their own fields: Role Requirements, Desired Qualifications and Skill Keywords. This corpus is the ground-truth on the semantic matching model, as it allows the system to align the candidate profiles with the employer requirements depending on the circumstances and not on the quantity of favored keywords.

C. Pre trained Embeddings The system relies on the already trained language embeddings which are BERT and Sentence BERT (S BERT) further adjusted with the above-discussed datasets. This is one of the ways in which transfer learning approach significantly increases the capacity of the model to perceive context, learners are able to distinguish, say, between Java the coffee and Java the programming language, thus, making it better at ranking these candidates and also generating interview questions.

D. Communication Templates To enable the automated notification system, a repertoire of synthetic HR email templates had been made. Standards of acceptance, rejection, and follow up communications are present in the dataset. Such templates are designed in such a way to test the variable insertion logic of the system, such that automated emails have a professional tone but have the ability to properly fill in candidate specific details.

E. Data Security and Privacy All datasets are extensively

sanitized before ingestion, to achieve the regulation of Personally Identifiable Information (PII) in accordance with the standards of data privacy. The anonymized data is kept on the secure clouds (AWS S3 and MongoDB Atlas). The system is known to enforce strict access control policies with data at rest encryption using industry standard AES 256 which guarantees the platform high levels of trust and transparency.

IV. RESULTS AND DISCUSSION

In this part, the results of the OptiHire framework are compared to the currently existing baseline methodologies, the importance of the results will be interpreted, and the shortcomings present in the system will be discussed candidly.

A. Comparative Analysis with Baselines To demonstrate the effectiveness of the proposed Semantic Similarity Engine, we compared OptiHire to two usual baselines, which included a representative Boolean Search system (Keyword Matching) and Word2Vec based probabilistic-based system.

According to Table I below, the Boolean method is very accurate although much less recalls, due to its inability to recognize qualified candidates when using synonyms (e.g., ReactJS vs. React.js). Although the Word2Vec model can be better applied to this as it also takes into account local context, it cannot cope with polysemy-words which have many different meanings depending on non-linguistic factors. OptiHire with the help of Sentence BERT shows a significant increase in F1 Score. This indicates that attention mechanism embedded in the transformer models is able to effectively extract long range dependencies of vectors in unstructured resume that are absent in previous embedding models.

TABLE I. PERFORMANCE OF PARSING METHODOLOGIES

Method	Precision	Recall	F1 Score	Avg. Time
Boolean (Keyword)	0.82	0.55	0.65	0.05
Word2Vec (Static)	0.76	0.72	0.74	0.12
OptiHire (S-BERT)	0.89	0.84	0.86	0.35

B. Major Benefits of the Proposed Architecture. Contextual Disambiguation: The main benefit of the fine tuned BERT model is that it can differentiate context. Considering Java as an example, the system can differentiate between Java at the programming skill section of the resume and Java at the non technical section of the resume which means a place or coffee.

Elimination of First Pass Bias: The standard method of manual screening can be prejudiced against the applicant by some unconscious bias over formatting or educational background. Allowing the input to be standardized through NER and scoring uses an objective assessment in mathematics with vectors similarity occurring before human intervention by OptiHire.

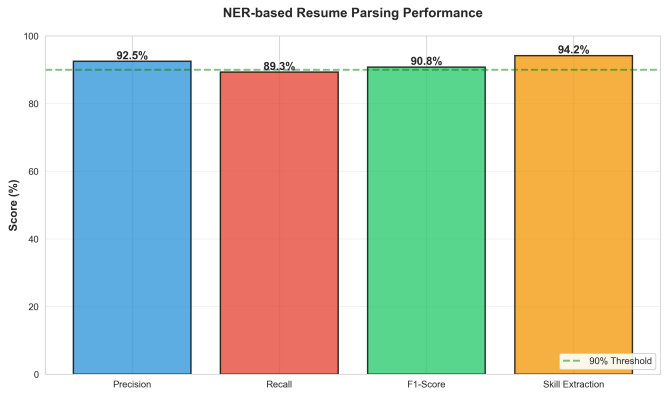

Fig. 2. NER-based Resume Parsing Performance

Important Metrics: Precision (92.5), Recall (89.3), F1-Score (90.8) and general Skill Extraction Accuracy (94.2).

Comparison of Method Performance. This compares different technical methods, including Boolean Search and Word2Vec, to the training of the S-BERT model in OptiHire to prove the better semantic match functions of the system.
Precise Comparisons, Recall, and F1-Score: The values of comparative metrics of various algorithms.

Finding Distribution of Semantic Matching Scores.

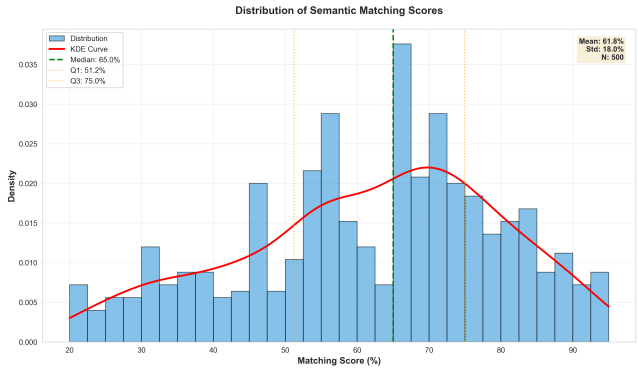

Fig. 3. Distribution of Semantic Matching Scores

Shows how the scores of the candidates are distributed in the talent pool giving the recruiter a statistical picture of where most of the applicants have their fit in the job.
Important measurements Mean (61.8%), Standard Deviation (18.0%), and Median (65.0%) equal scores of 500 applicants.

Feature Ratings Radar

The holistic radar chart that shows the perceived strengths of the users regarding many features that signifies the system excels in features like Semantic Matching and XAI Score Breakdowns.
Most Important Metrics: User ratings (1-5) of the core features such as Resume Parsing, Application Tracking, and

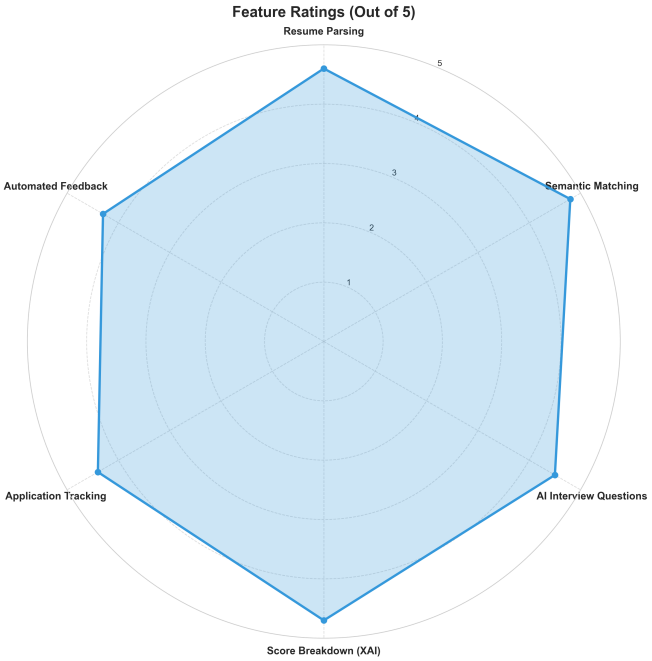

Fig. 4. Feature Ratings Radar

Automated Feedback.

V. CONCLUSION AND FUTURE SCOPE

The present paper introduced the OptimHire recruitment model that uses fine configured models of BERT and Generative AI to overcome the inefficiency of existing talent acquisition pipeline. The system can match semantic vectors by changing the manual keyword filtering with semantic vector matching (Sentence BERT), so that the system can gain a more profound and context sensitive knowledge of applicant profiles and thus filling the gap between applicant and job analysis. The experimental findings prove the existence of substantial screening latency reduction and the accuracy of candidate job matching brought out by the platform. Moreover, Explainable AI (XAI) will be integrated into the process to make the scoring process more transparent, which will make applicants more confident.

Future Scope Although the current version is performing well, in the future, the development will be based on three aspects:

Multimodal Assessment: Adding a Video Interview Module based on using WebRTC to allow the live coding sessions and behavior analysis.

State-of-the-art Bias minimization: Build Fairness-Aware ML models This aims to protect against linguistic bias in job descriptions and resume processing to facilitate fair jobs hiring.

Collaborative Ecosystems: Technical evaluations can be made easier by integrating real time technical assessment tools directly into the interview interface, e.g., collaborative code editors (e.g., revolving around VS Code Live Share).

REFERENCES

[1] V. M. Marciano, "The Origins and Development of Human Resource Management," *Academy of Management Proceedings*, vol. 1995, no. 1, pp. 223–227, 1995.

[2] R. B. Ewen, P. C. Smith, and C. L. Hulin, "An Empirical Test of the Herzberg Two-Factor Theory," *Journal of Applied Psychology*, vol. 50, no. 6, p. 544, 1966.

[3] E. L. Deci and R. M. Ryan, "Self-Determination Theory: When Mind Mediates Behavior," *The Journal of Mind and Behavior*, pp. 33–43, 1980.

[4] J. I. Tsujii, "Computational Linguistics and Natural Language Processing," in *International Conference on Intelligent Text Processing and Computational Linguistics*, Springer, Berlin, Heidelberg, pp. 52–67, 2011.

[5] B. Kitchenham, "Procedures for Performing Systematic Reviews," Joint Technical Report, Software Engineering Group, Keele University and Empirical Software Engineering National ICT Australia Ltd, 2004.

[6] B. Kitchenham, O. P. Brereton, D. Budgen, M. Turner, J. Bailey, and S. Linkman, "Systematic Literature Reviews in Software Engineering–A Systematic Literature Review," *Information and Software Technology*, vol. 51, no. 1, pp. 7–15, 2009.

[7] O. Filatova, "More Than a Word Cloud," *Tesol Journal*, vol. 7, no. 2, pp. 438–448, 2016.

[8] K. Yu, G. Guan, and M. Zhou, "Resume Information Extraction with Cascaded Hybrid Model," in *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL'05)*, pp. 499–506, 2005.

[9] F. Sebastiani, "Machine Learning in Automated Text Categorization," *ACM Computing Surveys*, vol. 34, no. 1, pp. 1–47, 2002.

[10] S. Katz, "Estimation of Probabilities from Sparse Data for the Language Model Component of a Speech Recognizer," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 35, no. 3, pp. 400–401, 1987.

[11] W. A. Gale and G. Sampson, "Good-Turing Frequency Estimation Without Tears," *Journal of Quantitative Linguistics*, vol. 2, no. 3, pp. 217–237, 1995.

[12] A. Z. Broder, "Identifying and Filtering Near-Duplicate Documents," in *Annual Symposium on Combinatorial Pattern Matching*, Springer, Berlin, Heidelberg, pp. 1–10, 2000.

[13] J. Lafferty, A. McCallum, and F. C. Pereira, "Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data," in *Proceedings of the 18th International Conference on Machine Learning*, 2001.

[14] A. Singh, C. Rose, K. Visweswariah, V. Chenthamarakshan, and N. Kambhatla, "PROSPECT: A System for Screening Candidates for Recruitment," in *Proceedings of the 19th ACM International Conference on Information and Knowledge Management*, pp. 659–668, 2010.

[15] K. Satheesh, A. Jahnavi, L. Iswarya, K. Ayesha, G. Bhanusekhar, and K. Hanisha, "Resume Ranking based on Job Description using SpaCy NER Model," 2020.

[16] S. Maheshwari and P. K. Reddy, "Discovering Special Product Features for Improving the Process of Product Selection in E-commerce Environment," in *Proceedings of the 11th International Conference on Electronic Commerce*, pp. 47–56, 2009.

[17] S. Maheshwari, A. Sainani, and P. K. Reddy, "An Approach to Extract Special Skills to Improve the Performance of Resume Selection," in *International Workshop on Databases in Networked Information Systems*, Springer, Berlin, Heidelberg, pp. 256–273, 2010.

[18] E. S. Chifu, V. R. Chifu, I. Popa, and I. Salomie, "A System for Detecting Professional Skills from Resumes Written in Natural Language," in *2017 13th IEEE International Conference on Intelligent Computer Communication and Processing (ICCP)*, pp. 189–196, 2017.

[19] D. Çelik and A. Elçi, "An Ontology-Based Information Extraction Approach for Résumés," in *Joint International Conference on Pervasive Computing and the Networked World*, Springer, Berlin, Heidelberg, pp. 165–179, 2012.

[20] V. Senthil Kumaran and A. Sankar, "Towards an Automated System for Intelligent Screening of Candidates for Recruitment Using Ontology Mapping (EXPERT)," *International Journal of Metadata, Semantics and Ontologies*, vol. 8, no. 1, pp. 56–64, 2013.

[21] A. M. Abirami, A. Askarunisa, R. Sangeetha, C. Padmavathi, and M. Priya, "Ontology Based Ranking of Documents Using Graph Databases: A Big Data Approach," 2017.

[22] A. Mohamed, W. Bagawathinathan, U. Iqbal, S. Shamrath, and A. Jayakody, "Smart Talents Recruiter-Resume Ranking and Recommendation System," in *2018 IEEE International Conference on Information and Automation for Sustainability (ICIAfS)*, pp. 1–5, 2018.

[23] U. Bojārs and J. G. Breslin, "ResumeRDF: Expressing Skill Information on the Semantic Web," in *1st International ExpertFinder Workshop*, 2007.

[24] D. Chandola, A. Garg, A. Maurya, and A. Kushwaha, "Online Resume Parsing System Using Text Analytics," *Journal of Multi Disciplinary Engineering Technologies (JMDET)*, vol. 9, 2015.

[25] S. K. Kopparapu, "Automatic Extraction of Usable Information from Unstructured Resumes to Aid Search," in *2010 IEEE International Conference on Progress in Informatics and Computing*, vol. 1, pp. 99–103, 2010.

[26] M. Thelen and E. Riloff, "A Bootstrapping Method for Learning Semantic Lexicons Using Extraction Pattern Contexts," in *Proceedings of the 2002 Conference on Empirical Methods in Natural Language Processing (EMNLP 2002)*, pp. 214–221, 2002.

[27] S. Pawar, R. Srivastava, and G. K. Palshikar, "Automatic Gazette Creation for Named Entity Recognition and Application to Resume Processing," in *Proceedings of the 5th ACM COMPUTE Conference: Intelligent & Scalable System Technologies*, pp. 1–7, 2012.

[28] R. Farkas, A. Dobó, Z. Kurai, I. Miklós, Á. Nagy, V. Vincze, and J. Zsibrita, "Information Extraction from Hungarian, English and German CVs for a Career Portal," in *Mining Intelligence and Knowledge Exploration*, Springer, Cham, pp. 333–341, 2014.

[29] J. Chen, Z. Niu, and H. Fu, "A Novel Knowledge Extraction Framework for Resumes Based on Text Classifier," in *International Conference on Web-Age Information Management*, Springer, Cham, pp. 540–543, 2015.

[30] M. Tikhonova and A. Gavrishchuk, "NLP Methods for Automatic Candidate's CV Segmentation," in *2019 International Conference on Engineering and Telecommunication (EnT)*, pp. 1–5, 2019.